

Inclusion–Exclusion

COMP2711 Discrete Mathematical Tools

Mingxun Zhou (mingxunz@ust.hk)

Example: Club memberships

Among **40 classmates**, 10 play badminton (A) and 20 go hiking (B).

How many students do neither activity?

Example: Club memberships

Among **40 classmates**, 10 play badminton (A) and 20 go hiking (B).

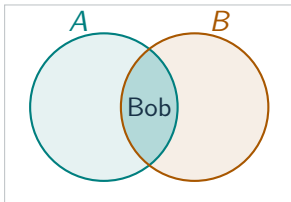
How many students do neither activity?

Is it just $40 - 10 - 20 = \mathbf{10}$?

Example: Club memberships

Among **40 classmates**, 10 play badminton (A) and 20 go hiking (B).

How many students do neither activity?



Not necessarily. Bob can do both.

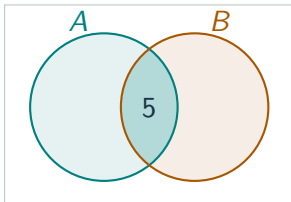
Subtracting 10 and 20 subtracts Bob twice.

We need the overlap count.

Example: Club memberships

Among **40 classmates**, 10 play badminton (A) and 20 go hiking (B).

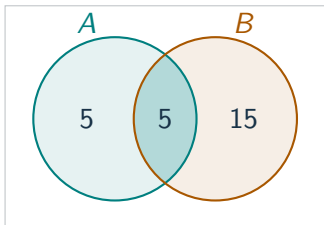
How many students do neither activity?



Now given: exactly 5 students do both.

How many students do neither activity?

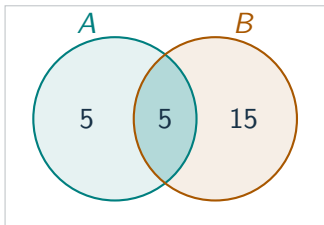
Solution: Count each student once



A: badminton B: hiking

$10 + 20$ counts the 5 students in both groups **twice**.

Solution: Count each student once

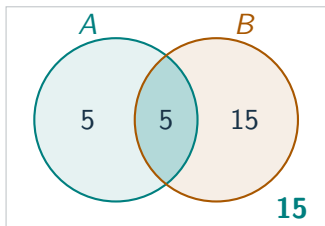


A: badminton B: hiking

$10 + 20$ counts the 5 students in both groups **twice**.

At least one activity: $10 + 20 - 5 = 25$.

Solution: Count each student once



A: badminton B: hiking

$10 + 20$ counts the 5 students in both groups **twice**.

At least one activity: $10 + 20 - 5 = 25$.

Neither activity: $40 - 25 = \mathbf{15}$.

Lecture structure

1. **Overlapping sets** Count each element once.

Lecture structure

1. **Overlapping sets** Count each element once.
2. **Applications** Interpret each intersection, then count it.

SECTION 1

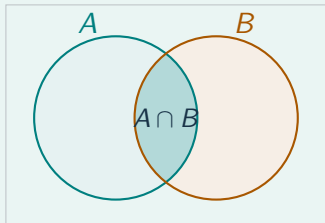
Counting overlapping sets

Two-set inclusion–exclusion

MUST UNDERSTAND

For finite sets A, B ,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$



Adding $|A| + |B|$ counts the overlap **twice**.

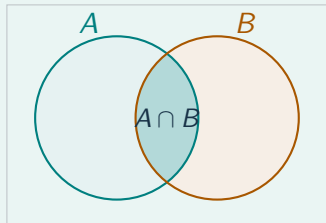
Subtract $|A \cap B|$ once, so every element is counted **once**.

Two-set inclusion–exclusion

MUST UNDERSTAND

For finite sets A, B ,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$



Adding $|A| + |B|$ counts the overlap **twice**.

Subtract $|A \cap B|$ once, so every element is counted **once**.

If $A \cap B = \emptyset$, this is the sum rule for disjoint sets.

Example: Three clubs

A new survey of **40 students** gives these results:

Play badminton	22
Go hiking	18
Play board games	15
Badminton and hiking	8
Badminton and board games	7
Hiking and board games	6
All three activities	3

Each pair count **includes** students who do all three.

How many students do at least one activity?

Three clubs: Set notation

Let A, B, C be the badminton, hiking, and board-games groups.

In words	In set notation
22 play badminton	
18 go hiking	
15 play board games	
8 do badminton and hiking	
7 do badminton and board games	
6 do hiking and board games	
3 do all three	
How many do at least one ?	

Three clubs: Set notation

Let A, B, C be the badminton, hiking, and board-games groups.

In words	In set notation
22 play badminton	$ A = 22$
18 go hiking	$ B = 18$
15 play board games	$ C = 15$
8 do badminton and hiking	
7 do badminton and board games	
6 do hiking and board games	
3 do all three	
How many do at least one ?	

Three clubs: Set notation

Let A, B, C be the badminton, hiking, and board-games groups.

In words	In set notation
22 play badminton	$ A = 22$
18 go hiking	$ B = 18$
15 play board games	$ C = 15$
8 do badminton and hiking	$ A \cap B = 8$
7 do badminton and board games	$ A \cap C = 7$
6 do hiking and board games	$ B \cap C = 6$
3 do all three	

How many do **at least one**?

Three clubs: Set notation

Let A, B, C be the badminton, hiking, and board-games groups.

In words	In set notation
22 play badminton	$ A = 22$
18 go hiking	$ B = 18$
15 play board games	$ C = 15$
8 do badminton and hiking	$ A \cap B = 8$
7 do badminton and board games	$ A \cap C = 7$
6 do hiking and board games	$ B \cap C = 6$
3 do all three	$ A \cap B \cap C = 3$

How many do **at least one**?

Three clubs: Set notation

Let A, B, C be the badminton, hiking, and board-games groups.

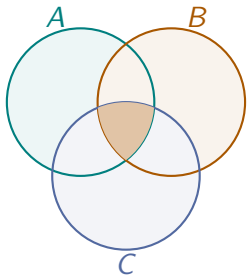
In words	In set notation
22 play badminton	$ A = 22$
18 go hiking	$ B = 18$
15 play board games	$ C = 15$
8 do badminton and hiking	$ A \cap B = 8$
7 do badminton and board games	$ A \cap C = 7$
6 do hiking and board games	$ B \cap C = 6$
3 do all three	$ A \cap B \cap C = 3$
How many do at least one ?	$ A \cup B \cup C = ?$

Solution 1: All three activities

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



Start with the **triple intersection**.

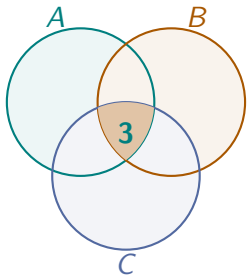
$$|A \cap B \cap C| = 3.$$

Solution 1: All three activities

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



Start with the **triple intersection**.

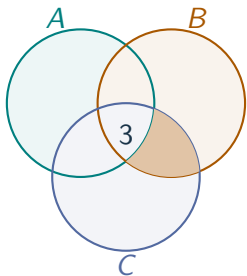
$$|A \cap B \cap C| = 3.$$

Solution 1: Exactly two activities

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



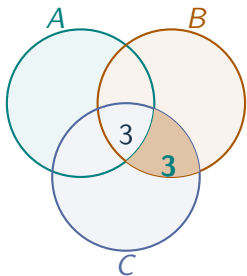
Of the 6 who hike and play board games, 3 also play badminton.

Solution 1: Exactly two activities

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



Of the 6 who hike and play board games, 3 also play badminton.

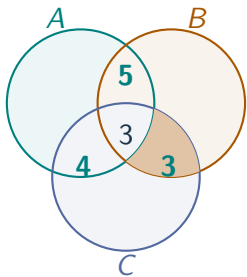
$$|(B \cap C) \setminus A| = 6 - 3 = \mathbf{3}.$$

Solution 1: Exactly two activities

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



Of the 6 who hike and play board games, 3 also play badminton.

$$|(B \cap C) \setminus A| = 6 - 3 = \mathbf{3}.$$

Similarly, excluding the third activity:

$$|(A \cap B) \setminus C| = 8 - 3 = \mathbf{5},$$

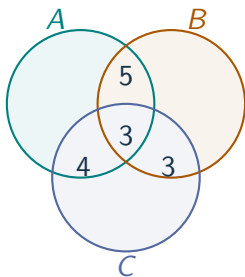
$$|(A \cap C) \setminus B| = 7 - 3 = \mathbf{4}.$$

Solution 1: Only one activity

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



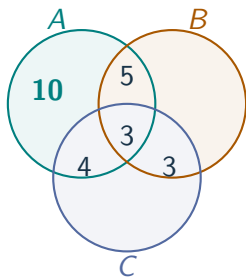
Subtract the **already filled regions** inside each club.

Solution 1: Only one activity

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



Subtract the **already filled regions** inside each club.

Badminton only:

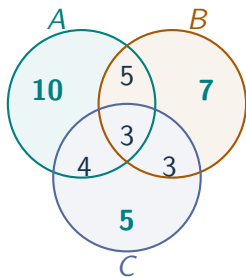
$$22 - 5 - 4 - 3 = \mathbf{10}.$$

Solution 1: Only one activity

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



Subtract the **already filled regions** inside each club.

Badminton only:

$$22 - 5 - 4 - 3 = \mathbf{10}.$$

$$\text{Hiking only: } 18 - 5 - 3 - 3 = \mathbf{7}.$$

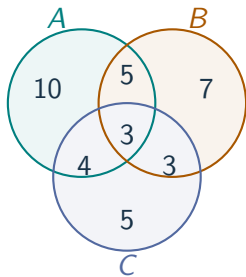
$$\text{Board games only: } 15 - 4 - 3 - 3 = \mathbf{5}.$$

Solution 1: Add the regions

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



These seven regions are **disjoint**.

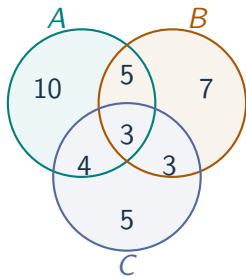
$$\begin{aligned} |A \cup B \cup C| &= 10 + 7 + 5 \\ &\quad + 5 + 4 + 3 + 3 = \mathbf{37}. \end{aligned}$$

Solution 1: Add the regions

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



These seven regions are **disjoint**.

$$\begin{aligned} |A \cup B \cup C| &= 10 + 7 + 5 \\ &\quad + 5 + 4 + 3 + 3 = \mathbf{37}. \end{aligned}$$

$$\text{None: } 40 - 37 = 3.$$

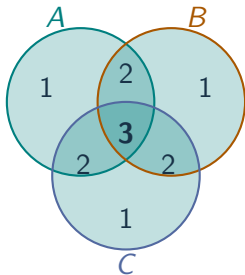
$$\text{Exactly two: } 5 + 4 + 3 = 12.$$

Solution 2: Counting and correcting

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



1. Add the three set sizes.

$$|A| + |B| + |C| = 22 + 18 + 15 = 55.$$

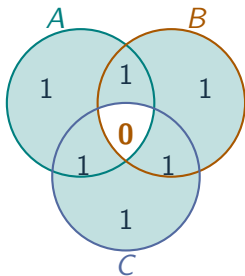
Times each student is counted

Solution 2: Counting and correcting

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



Times each student is counted

Counting overlapping sets

2. Subtract all pair intersections.

$$-|A \cap B| - |A \cap C| - |B \cap C|$$

$$55 - 8 - 7 - 6 = 34.$$

Pair-only regions: counted once.

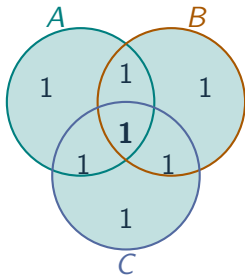
Center: $3 - 3 = 0$, instead of 1.

Solution 2: Counting and correcting

40 classmates: A badminton, B hiking, C board games.

$$|A| = 22, \quad |B| = 18, \quad |C| = 15.$$

$$|A \cap B| = 8, \quad |A \cap C| = 7, \quad |B \cap C| = 6, \quad |A \cap B \cap C| = 3.$$



Times each student is counted

Counting overlapping sets

3. Add the triple intersection back.

$$+|A \cap B \cap C| = +3.$$

Every student in the union now counts once.

$$|A \cup B \cup C| = 34 + 3 = \mathbf{37}.$$

Three-set inclusion–exclusion

For finite sets A, B, C ,

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|. \end{aligned}$$

General inclusion–exclusion principle

For finite sets $A_1, \dots, A_n$, where $n \geq 1$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} |A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}|.$$

General inclusion–exclusion principle

For finite sets $A_1, \dots, A_n$, where $n \geq 1$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} |A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}|.$$

Level k : intersections of k distinct sets.

General inclusion–exclusion principle

For finite sets $A_1, \dots, A_n$, where $n \geq 1$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} |A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}|.$$

Level k : intersections of k distinct sets.

Odd levels (1, 3, 5, ...): **add** all intersection counts.

Even levels (2, 4, 6, ...): **subtract** them.

Proof: Count one element

Suppose x belongs to exactly $r \geq 1$ of the sets.

Terms containing x	Choose from those r sets	Contribution
Single sets	1 set	$+\binom{r}{1}$

Proof: Count one element

Suppose x belongs to exactly $r \geq 1$ of the sets.

Terms containing x	Choose from those r sets	Contribution
Single sets	1 set	$+\binom{r}{1}$
Pair intersections	2 sets	$-\binom{r}{2}$
Triple intersections	3 sets	$+\binom{r}{3}$
$\vdots$	$\vdots$	$\vdots$

Proof: Count one element

Suppose x belongs to exactly $r \geq 1$ of the sets.

Terms containing x	Choose from those r sets	Contribution
Single sets	1 set	$+\binom{r}{1}$
Pair intersections	2 sets	$-\binom{r}{2}$
Triple intersections	3 sets	$+\binom{r}{3}$
$\vdots$	$\vdots$	$\vdots$

We need
$$\binom{r}{1} - \binom{r}{2} + \cdots + (-1)^{r+1} \binom{r}{r} = 1.$$

Proof: Use the binomial theorem

Set $a = 1$ and $b = -1$, with $r \geq 1$.

Recall: Binomial theorem

$$(a + b)^r = \sum_{k=0}^r \binom{r}{k} a^{r-k} b^k$$

$$0 = (1 - 1)^r = 1 - \binom{r}{1} + \binom{r}{2} - \cdots + (-1)^r \binom{r}{r}.$$

Proof: Use the binomial theorem

Set $a = 1$ and $b = -1$, with $r \geq 1$.

Recall: Binomial theorem

$$(a + b)^r = \sum_{k=0}^r \binom{r}{k} a^{r-k} b^k$$

$$0 = (1 - 1)^r = 1 - \binom{r}{1} + \binom{r}{2} - \cdots + (-1)^r \binom{r}{r}.$$

$$\binom{r}{1} - \binom{r}{2} + \cdots + (-1)^{r+1} \binom{r}{r} = \mathbf{1}.$$

Proof: Use the binomial theorem

Set $a = 1$ and $b = -1$, with $r \geq 1$.

Recall: Binomial theorem

$$(a + b)^r = \sum_{k=0}^r \binom{r}{k} a^{r-k} b^k$$

$$0 = (1 - 1)^r = 1 - \binom{r}{1} + \binom{r}{2} - \cdots + (-1)^r \binom{r}{r}.$$

$$\binom{r}{1} - \binom{r}{2} + \cdots + (-1)^{r+1} \binom{r}{r} = \mathbf{1}.$$

Every element in the union is counted once.

An element outside every set is counted zero times.

When to use inclusion–exclusion: Case 1

Inclusion–exclusion turns an **OR count** into **AND counts**.

Target: Count x such that

$x \in A$ **or** $x \in B$ **or** $x \in C, \dots$

When to use inclusion–exclusion: Case 1

Inclusion–exclusion turns an **OR count** into **AND counts**.

Target: Count x such that

$x \in A$ **or** $x \in B$ **or** $x \in C, \dots$

$\implies$

Turns into: Count x such that

$x \in A$ **and** $x \in B$ **and** $x \in C, \dots$

Count each single set, each pairwise intersection, each triple intersection, and so on.

When to use inclusion–exclusion: Case 2

Complement case

Count x satisfying **none** of conditions $A, B, C, \dots$

Let U be all candidates; $A, B, C, \dots$ are the bad sets.

When to use inclusion–exclusion: Case 2

Complement case

Count x satisfying **none** of conditions $A, B, C, \dots$

Let U be all candidates; $A, B, C, \dots$ are the bad sets.

$$\# \text{valid} = |U| - |A \cup B \cup C \cup \dots|.$$

SECTION 2

Applications

Application 1: Divisibility

Consider the integers $1, 2, \dots, 100$.

How many are divisible by at least one of 2, 3, 5?

8	9	10	30
divisible by 2	divisible by 3	divisible by 2, 5	divisible by 2, 3, 5

Solution: Count the intersections

Let A_d contain the multiples of d from 1 through 100.

$$|A_d| = \left\lfloor \frac{100}{d} \right\rfloor.$$

Sets	Multiples of	Counts
A_2, A_3, A_5	2, 3, 5	50, 33, 20

Solution: Count the intersections

Let A_d contain the multiples of d from 1 through 100.

$$|A_d| = \left\lfloor \frac{100}{d} \right\rfloor.$$

Sets	Multiples of	Counts
A_2, A_3, A_5	2, 3, 5	50, 33, 20
$A_2 \cap A_3, A_2 \cap A_5, A_3 \cap A_5$	6, 10, 15	16, 10, 6

Common multiples of a, b are multiples of $\text{lcm}(a, b)$.

Solution: Count the intersections

Let A_d contain the multiples of d from 1 through 100.

$$|A_d| = \left\lfloor \frac{100}{d} \right\rfloor.$$

Sets	Multiples of	Counts
A_2, A_3, A_5	2, 3, 5	50, 33, 20
$A_2 \cap A_3, A_2 \cap A_5, A_3 \cap A_5$	6, 10, 15	16, 10, 6
$A_2 \cap A_3 \cap A_5$	30	3

Common multiples of a, b are multiples of $\text{lcm}(a, b)$.

Solution: Divisibility

$$\begin{aligned} |A_2 \cup A_3 \cup A_5| &= \underbrace{50 + 33 + 20}_{\text{single sets}} - \underbrace{(16 + 10 + 6)}_{\text{pair intersections}} + \underbrace{3}_{\text{triple}} \\ &= 74. \end{aligned}$$

Solution: Divisibility

$$\begin{aligned} |A_2 \cup A_3 \cup A_5| &= \underbrace{50 + 33 + 20}_{\text{single sets}} - \underbrace{(16 + 10 + 6)}_{\text{pair intersections}} + \underbrace{3}_{\text{triple}} \\ &= 74. \end{aligned}$$

Divisible by none: $100 - 74 = 26$.

Practice: Shared factors

How many integers from 1 through 100 are divisible by **6 or 10**?

Practice: Shared factors

How many integers from 1 through 100 are divisible by **6 or 10**?

Both means a multiple of $\text{lcm}(6, 10) = 30$, **not 60**.

Practice: Shared factors

How many integers from 1 through 100 are divisible by **6 or 10**?

Both means a multiple of $\text{lcm}(6, 10) = 30$, **not 60**.

$$\left\lfloor \frac{100}{6} \right\rfloor + \left\lfloor \frac{100}{10} \right\rfloor - \left\lfloor \frac{100}{30} \right\rfloor = 16 + 10 - 3 = \mathbf{23}.$$

Application 2: Digit codes

A code has **five positions**, each containing 0, 1, or 2. Repetition and leading zeros are allowed.

0	0	1	2	2
0	0	1	1	0

all three digits appear

digit 2 is missing

How many codes contain every digit at least once?

A tempting count

Choose one position for each required digit, then fill the other positions.

Shaded: chosen first. Gray: filled afterward.

0	0	1	2	2
---	---	---	---	---

Same final code: 00122.

A tempting count

Choose one position for each required digit, then fill the other positions.

Shaded: chosen first. Gray: filled afterward.

0	0	1	2	2
0	0	1	2	2
0	0	1	2	2
0	0	1	2	2

Same final code: 00122.

$2 \cdot 1 \cdot 2 = 4$ descriptions.

But 00012 has only

$3 \cdot 1 \cdot 1 = 3$ descriptions.

A tempting count

Choose one position for each required digit, then fill the other positions.

Shaded: chosen first. Gray: filled afterward.

0	0	1	2	2
0	0	1	2	2
0	0	1	2	2
0	0	1	2	2

Same final code: 00122.

$2 \cdot 1 \cdot 2 = 4$ descriptions.

But 00012 has only

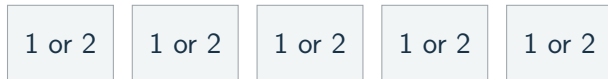
$3 \cdot 1 \cdot 1 = 3$ descriptions.

No uniform overcount. Instead, count codes with a **missing digit**.

Codes: All candidates and bad sets

All candidates: 3^5 codes.

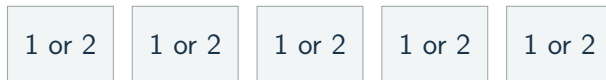
Let E_0 be the codes missing digit 0.



Codes: All candidates and bad sets

All candidates: 3^5 codes.

Let E_0 be the codes missing digit 0.

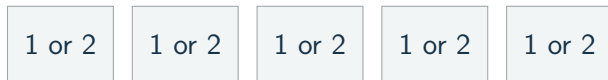


$$|E_0| = |E_1| = |E_2| = 2^5.$$

Codes: All candidates and bad sets

All candidates: 3^5 codes.

Let E_0 be the codes missing digit 0.



$$|E_0| = |E_1| = |E_2| = 2^5.$$

These sets overlap: 22222 misses both 0 and 1.

Codes: Intersections

$E_0 \cap E_1$: **neither 0 nor 1**

2	2	2	2	2
---	---	---	---	---

$$|E_0 \cap E_1| = 1.$$

Codes: Intersections

$E_0 \cap E_1$: **neither 0 nor 1**

2	2	2	2	2
---	---	---	---	---

$$|E_0 \cap E_1| = 1.$$

$E_0 \cap E_1 \cap E_2$

No available digits:

$$|E_0 \cap E_1 \cap E_2| = 0.$$

Codes: Intersections

$E_0 \cap E_1$: **neither 0 nor 1**

2	2	2	2	2
---	---	---	---	---

$$|E_0 \cap E_1| = 1.$$

$E_0 \cap E_1 \cap E_2$

No available digits:

$$|E_0 \cap E_1 \cap E_2| = 0.$$

An intersection forbids the **specified** digits; others may be missing too.

Codes: Inclusion–exclusion

Forbidden digits	Choices of digits	Codes per choice	Contribution
0	1	3^5	+243
1	$\binom{3}{1} = 3$	2^5	−96
2	$\binom{3}{2} = 3$	1^5	+3
3	1	0	0

Codes: Inclusion–exclusion

Forbidden digits	Choices of digits	Codes per choice	Contribution
0	1	3^5	+243
1	$\binom{3}{1} = 3$	2^5	−96
2	$\binom{3}{2} = 3$	1^5	+3
3	1	0	0

$$3^5 - \binom{3}{1} 2^5 + \binom{3}{2} 1^5 = 243 - 96 + 3 = \mathbf{150}.$$

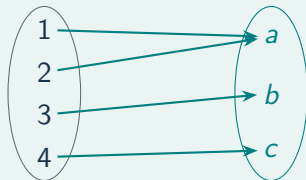
Recall: Counting all functions

Each input maps to any element of B .

$$\# \text{functions} = 3^4 = 81.$$

For $|A| = m$ and $|B| = n$: n^m **functions**.

$$A = \{1, 2, 3, 4\} \quad B = \{a, b, c\}$$

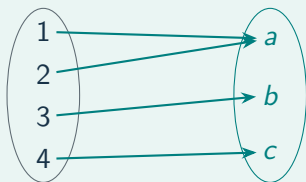


One possible function

Application 3: Surjective functions

$f : A \rightarrow B$ is surjective when
every element of B is reached.

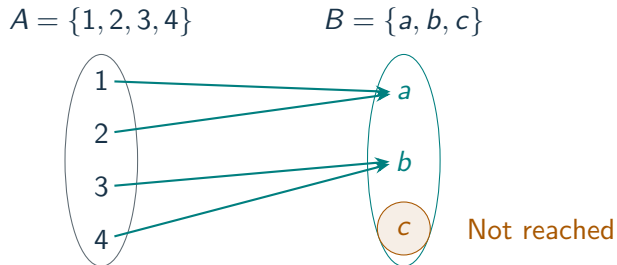
$$A = \{1, 2, 3, 4\} \quad B = \{a, b, c\}$$



For $|A| = m \geq 1$ and $|B| = n \geq 1$:

How many functions $f : A \rightarrow B$ reach every output?

Surjective functions: Bad sets and intersections



F_c : functions that never output c .

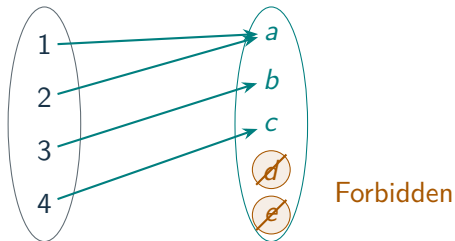
Surjective functions: Missing k outputs

Forbid outputs d and e .

Map each input into $\{a, b, c\}$.

$$\# \text{functions} = 3^4.$$

$$A = \{1, 2, 3, 4\} \quad B = \{a, b, c, d, e\}$$



$$\# \text{functions missing } k \text{ specified outputs} = (n - k)^m.$$

Surjective functions: Inclusion–exclusion

A function is not surjective if it misses b_1 **or** b_2 **or** $\cdots$ **or** b_n .

Surjective functions: Inclusion–exclusion

A function is not surjective if it misses b_1 **or** b_2 **or** $\dots$ **or** b_n .

Missing k specified outputs together: $(n - k)^m$ functions.

Choose the forbidden outputs: $\binom{n}{k}$ possibilities.

$$\# \text{surjective functions} = n^m - \binom{n}{1}(n-1)^m + \binom{n}{2}(n-2)^m - \dots$$

Surjective functions: Inclusion–exclusion

A function is not surjective if it misses b_1 **or** b_2 **or** $\dots$ **or** b_n .

Missing k specified outputs together: $(n - k)^m$ functions.

Choose the forbidden outputs: $\binom{n}{k}$ possibilities.

$$\# \text{surjective functions} = n^m - \binom{n}{1}(n-1)^m + \binom{n}{2}(n-2)^m - \dots$$

$$\# \text{surjective functions} = \sum_{k=0}^n (-1)^k \underbrace{\binom{n}{k}}_{\text{choose forbidden outputs}} \underbrace{(n-k)^m}_{\text{map to the remaining outputs}}.$$

If $m < n$, no surjective function exists.

Application 4: Derangements

Permute the numbers $1, 2, \dots, n$.

How many permutations have no number i in position i ?

Valid: $n = 4$

Position			
1	2	3	4
2	3	4	1

Invalid: $n = 4$

Position			
1	2	3	4
2	3	1	4

4 in position 4

Derangements: AND and OR conditions

Valid: 1 is not in position 1 **and** 2 is not in position 2
and $\dots$ **and** n is not in position n .

Derangements: AND and OR conditions

Valid: 1 is not in position 1 **and** 2 is not in position 2
and $\dots$ **and** n is not in position n .

Invalid: 1 is in position 1 **or** 2 is in position 2
or $\dots$ **or** n is in position n .

Derangements: AND and OR conditions

Valid: 1 is not in position 1 **and** 2 is not in position 2
and $\dots$ **and** n is not in position n .

Invalid: 1 is in position 1 **or** 2 is in position 2
or $\dots$ **or** n is in position n .

Count the invalid permutations using **inclusion–exclusion**.

$$\# \text{valid} = n! - \# \text{invalid}.$$

Derangements: Specified numbers in place

For $n = 4$, require 1 to be in position 1.

position 1 position 2 position 3 position 4

1

?

?

?

2, 3, 4: $3!$ ways

Derangements: Specified numbers in place

For $n = 4$, require 1 to be in position 1.

position 1 position 2 position 3 position 4

1

?

?

?

2, 3, 4: $3!$ ways

Fix positions 1 and 2: permute 3, 4 in $2!$ ways.

Derangements: Specified numbers in place

For $n = 4$, require 1 to be in position 1.

position 1 position 2 position 3 position 4

1

?

?

?

2, 3, 4: $3!$ ways

Fix positions 1 and 2: permute 3, 4 in $2!$ ways.

Require k specified numbers to be in place: $(n - k)!$ permutations.

Choose those k numbers: $\binom{n}{k}$ choices.

The other numbers **may also** be in their own positions.

Derangements: Inclusion–exclusion for $n = 4$

Specified numbers in place	Choose them	Permute the rest	Contribution
0	1	$4! = 24$	$+24$
1	$\binom{4}{1}$	$3! = 6$	-24
2	$\binom{4}{2}$	$2! = 2$	$+12$
3	$\binom{4}{3}$	$1! = 1$	-4
4	$\binom{4}{4}$	$0! = 1$	$+1$

Derangements: Inclusion–exclusion for $n = 4$

Specified numbers in place	Choose them	Permute the rest	Contribution
0	1	$4! = 24$	$+24$
1	$\binom{4}{1}$	$3! = 6$	-24
2	$\binom{4}{2}$	$2! = 2$	$+12$
3	$\binom{4}{3}$	$1! = 1$	-4
4	$\binom{4}{4}$	$0! = 1$	$+1$

$$24 - 24 + 12 - 4 + 1 = \mathbf{9}.$$

Derangements

A **derangement** is a permutation in which no object stays in its own position.

For $n \geq 1$, its count is

$$D_n = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)!.$$

Derangements

A **derangement** is a permutation in which no object stays in its own position.

For $n \geq 1$, its count is

$$D_n = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)!.$$

Since $\binom{n}{k} (n-k)! = n!/k!$,

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

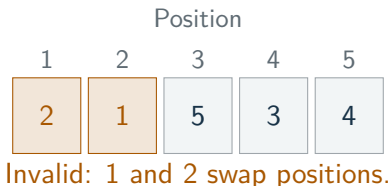
$D_0 = 1$ (the empty arrangement); $D_1 = 0$; $D_4 = 9$.

Challenge: Derangements with no pair swaps

Highly recommended exercise

Start with a derangement of $1, 2, \dots, n$.

What if we also require that **no two numbers swap their positions**?



How many derangements have no pair swaps? Use PIE.

Here $\pi(i)$ is the number in position i .

Full solution in the course note.

Challenge: Two types of bad events

Type A: Fixed position

Number i stays
in position i .



$$A_i : \pi(i) = i.$$

Type B: Pair swap

Position j contains k ;
position k contains j .



$$B_{jk} : \pi(j) = k \text{ and } \pi(k) = j \quad (j < k).$$

Challenge: Two types of bad events

Type A: Fixed position

Number i stays
in position i .



$$A_i : \pi(i) = i.$$

Type B: Pair swap

Position j contains k ;
position k contains j .



$$B_{jk} : \pi(j) = k \text{ and } \pi(k) = j \quad (j < k).$$

Key hint: PIE turns OR into AND counts.

$$A_i \cap B_{jk} : \pi(i) = i \text{ and } \pi(j) = k \text{ and } \pi(k) = j.$$

Can both events hold? How many permutations satisfy both?

Application 5: Integer solutions

Let n, r be nonnegative integers.

How many integer solutions satisfy

$$x_1 + x_2 + x_3 = n, \quad 0 \leq x_1, x_2, x_3 \leq r?$$

For example, take $n = 8$ and $r = 3$:

$(3, 3, 2)$ **Valid**: sum 8, each entry at most 3.

$(4, 3, 1)$ **Invalid**: the first entry exceeds 3.

Integer solutions: Ignore the upper bounds

First count all **nonnegative** solutions of $x_1 + x_2 + x_3 = n$.

Stars and bars: n stars and 2 bars.

$$n = 8: \quad \underbrace{\star \star \star}_{x_1 = 3} \mid \underbrace{\star \star \star}_{x_2 = 3} \mid \underbrace{\star \star}_{x_3 = 2}$$

$$\# \text{all candidates} = \binom{n+2}{2}.$$

Integer solutions: Ignore the upper bounds

First count all **nonnegative** solutions of $x_1 + x_2 + x_3 = n$.

Stars and bars: n stars and 2 bars.

$$n = 8: \quad \begin{array}{ccc|ccc|cc} \star & \star & \star & & \star & \star & \star & & \star & \star \\ \hline & & x_1 = 3 & & & x_2 = 3 & & & x_3 = 2 \end{array}$$

$$\# \text{all candidates} = \binom{n+2}{2}.$$

For $n = 8$: $\binom{10}{2} = 45$ candidates.

Some exceed r . **Subtract the invalid ones.**

Integer solutions: Remove the invalid solutions

Use **inclusion–exclusion** for the invalid conditions:

$$x_1 \geq r + 1 \quad \text{or} \quad x_2 \geq r + 1 \quad \text{or} \quad x_3 \geq r + 1.$$

Integer solutions: Remove the invalid solutions

Use **inclusion–exclusion** for the invalid conditions:

$$x_1 \geq r + 1 \quad \text{or} \quad x_2 \geq r + 1 \quad \text{or} \quad x_3 \geq r + 1.$$

Example intersection: $x_1 \geq r + 1$ **and** $x_2 \geq r + 1$, with $x_1 + x_2 + x_3 = n$.

Integer solutions: Remove the invalid solutions

Use **inclusion–exclusion** for the invalid conditions:

$$x_1 \geq r + 1 \quad \text{or} \quad x_2 \geq r + 1 \quad \text{or} \quad x_3 \geq r + 1.$$

Example intersection: $x_1 \geq r + 1$ **and** $x_2 \geq r + 1$, with $x_1 + x_2 + x_3 = n$.

Subtract $r + 1$ from each of the first two variables:

$$\underbrace{x_1 - (r + 1)}_{\geq 0} + \underbrace{x_2 - (r + 1)}_{\geq 0} + \underbrace{x_3}_{\geq 0} = n - 2(r + 1).$$

Integer solutions: Remove the invalid solutions

Use **inclusion–exclusion** for the invalid conditions:

$$x_1 \geq r + 1 \quad \text{or} \quad x_2 \geq r + 1 \quad \text{or} \quad x_3 \geq r + 1.$$

Example intersection: $x_1 \geq r + 1$ **and** $x_2 \geq r + 1$, with $x_1 + x_2 + x_3 = n$.

Subtract $r + 1$ from each of the first two variables:

$$\underbrace{x_1 - (r + 1)}_{\geq 0} + \underbrace{x_2 - (r + 1)}_{\geq 0} + \underbrace{x_3}_{\geq 0} = n - 2(r + 1).$$

Ordinary stars and bars again:

$$\binom{n - 2(r + 1) + 2}{2} = \binom{n - 2r}{2}.$$

Integer solutions: Inclusion–exclusion

All candidates – singles + pairs – triple.

Require	Count per choice	Choices
One chosen variable $\geq r + 1$	$\binom{n-r+1}{2}$	3
Two chosen variables $\geq r + 1$	$\binom{n-2r}{2}$	3
All three variables $\geq r + 1$	$\binom{n-3r-1}{2}$	1

Negative remaining sum: count 0 (use $\binom{a}{2} = 0$ for $a < 2$).

Integer solutions: Inclusion–exclusion

All candidates – singles + pairs – triple.

Require	Count per choice	Choices
One chosen variable $\geq r + 1$	$\binom{n-r+1}{2}$	3
Two chosen variables $\geq r + 1$	$\binom{n-2r}{2}$	3
All three variables $\geq r + 1$	$\binom{n-3r-1}{2}$	1

$$\# \text{valid} = \binom{n+2}{2} - 3 \binom{n-r+1}{2} + 3 \binom{n-2r}{2} - \binom{n-3r-1}{2}.$$

Negative remaining sum: count 0 (use $\binom{a}{2} = 0$ for $a < 2$).

Integer solutions: Inclusion–exclusion

All candidates – singles + pairs – triple.

Require	Count per choice	Choices
One chosen variable $\geq r + 1$	$\binom{n-r+1}{2}$	3
Two chosen variables $\geq r + 1$	$\binom{n-2r}{2}$	3
All three variables $\geq r + 1$	$\binom{n-3r-1}{2}$	1

$$\# \text{valid} = \binom{n+2}{2} - 3 \binom{n-r+1}{2} + 3 \binom{n-2r}{2} - \binom{n-3r-1}{2}.$$

Negative remaining sum: count 0 (use $\binom{a}{2} = 0$ for $a < 2$).

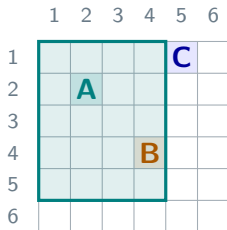
For $n = 8, r = 3$: $45 - 3 \cdot 15 + 3 \cdot 1 - 0 = \mathbf{3}$.

Application 6: Rectangles containing marked cells

Mark cells A, B, C in an $n \times n$ grid.

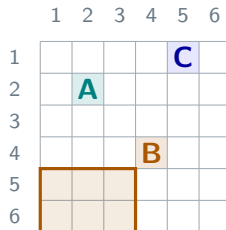
How many rectangles contain at least one whole marked cell?

Edges follow grid lines and stay inside the grid.



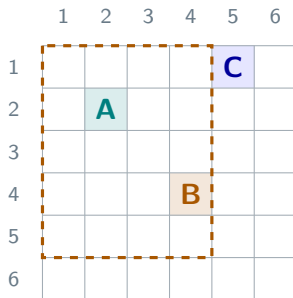
Valid: contains A and B .

This 6×6 example: $A = (2, 2)$, $B = (4, 4)$, $C = (1, 5)$.



Invalid: contains none.

Rectangles: Solution roadmap



One rectangle containing A and B .

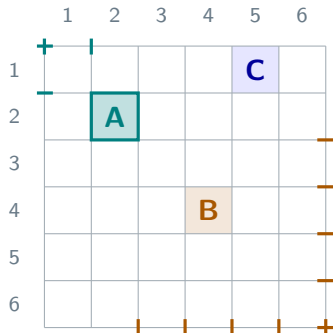
R_A : rectangles containing A .

Define R_B, R_C similarly.

Target: $|R_A \cup R_B \cup R_C|$.

1. Choose four boundaries.
2. Count singles, pairs, and the triple.
3. Apply inclusion–exclusion.

Rectangles containing one cell

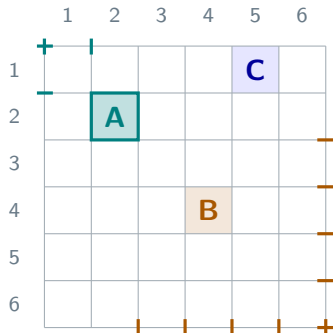


Top / left options
Bottom / right options

A is in row 2, column 2.

Boundary	Choices
Top	2
Bottom	5
Left	2
Right	5

Rectangles containing one cell



Top / left options

Bottom / right options

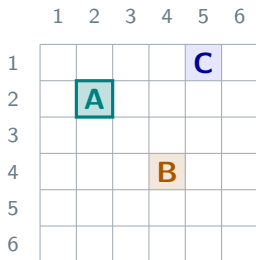
A is in row 2, column 2.

Boundary	Choices
Top	2
Bottom	5
Left	2
Right	5

$$|R_A| = 2 \cdot 5 \cdot 2 \cdot 5 = \mathbf{100}.$$

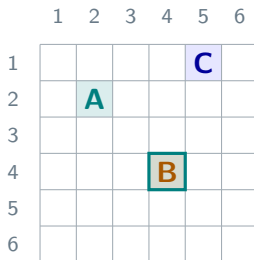
Each rectangle has a unique top, bottom, left, and right boundary.

Rectangles: The three single counts

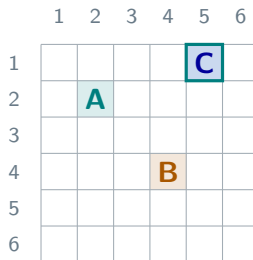


$$A = (2, 2)$$

$$|R_A| = 2 \cdot 5 \cdot 2 \cdot 5 = \mathbf{100}$$

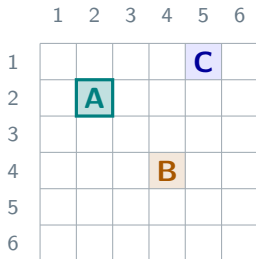


$$B = (4, 4)$$



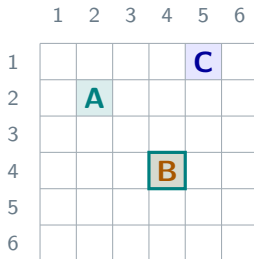
$$C = (1, 5)$$

Rectangles: The three single counts



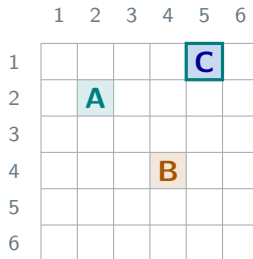
$$A = (2, 2)$$

$$|R_A| = 2 \cdot 5 \cdot 2 \cdot 5 = \mathbf{100}$$



$$B = (4, 4)$$

$$|R_B| = 4 \cdot 3 \cdot 4 \cdot 3 = \mathbf{144}$$

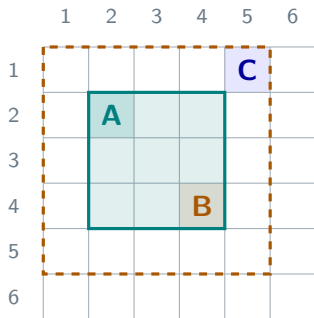


$$C = (1, 5)$$

$$|R_C| = 1 \cdot 6 \cdot 5 \cdot 2 = \mathbf{60}$$

Rectangles containing A and B

A rectangle contains A and B exactly when it contains their **bounding box**.



The box spans rows 2–4 and columns 2–4.

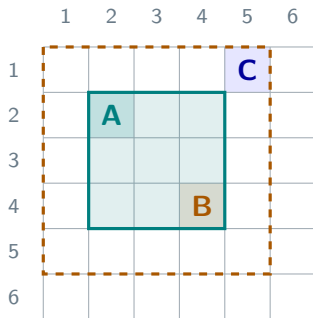
Top: 2 Bottom: 3

Left: 2 Right: 3

Solid: smallest enclosing box. Dashed: one containing rectangle.

Rectangles containing A and B

A rectangle contains A and B exactly when it contains their **bounding box**.



The box spans rows 2–4 and columns 2–4.

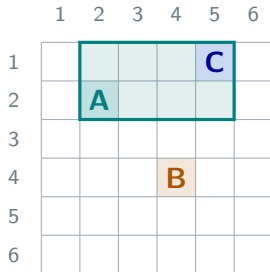
Top: 2 Bottom: 3

Left: 2 Right: 3

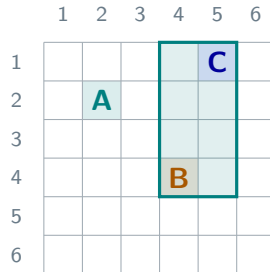
$$|R_A \cap R_B| = 2 \cdot 3 \cdot 2 \cdot 3 = \mathbf{36}.$$

Solid: smallest enclosing box. Dashed: one containing rectangle.

Rectangles: The other pair counts



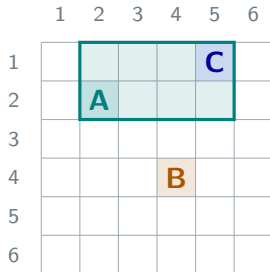
A, C: rows 1–2, columns 2–5.



B, C: rows 1–4, columns 4–5.

A pair count allows the rectangle to contain the third cell too.

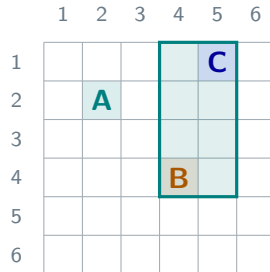
Rectangles: The other pair counts



A, C: rows 1–2, columns 2–5.

$$|R_A \cap R_C| = 1 \cdot 5 \cdot 2 \cdot 2 = \mathbf{20}.$$

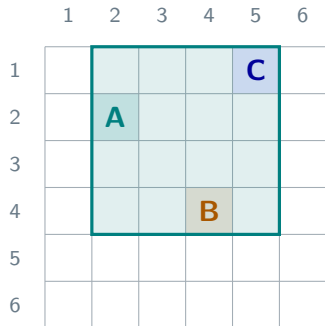
A pair count allows the rectangle to contain the third cell too.



B, C: rows 1–4, columns 4–5.

$$|R_B \cap R_C| = 1 \cdot 3 \cdot 4 \cdot 2 = \mathbf{24}.$$

Rectangles containing all three cells



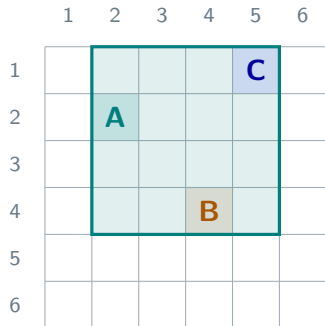
The bounding box spans

rows 1–4 and **columns 2–5**.

Top: 1 Bottom: 3

Left: 2 Right: 2

Rectangles containing all three cells



The bounding box spans

rows 1–4 and **columns 2–5**.

Top: 1 Bottom: 3

Left: 2 Right: 2

$$|R_A \cap R_B \cap R_C| = 1 \cdot 3 \cdot 2 \cdot 2 = \mathbf{12}.$$

Rectangles: Combine the seven counts

	Required cells	Rectangle counts
Add	A, B, C	100, 144, 60
Subtract	AB, AC, BC	36, 20, 24
Add	ABC	12

Rectangles: Combine the seven counts

	Required cells	Rectangle counts
Add	A, B, C	100, 144, 60
Subtract	AB, AC, BC	36, 20, 24
Add	ABC	12

$$\begin{aligned} |R_A \cup R_B \cup R_C| &= 100 + 144 + 60 - 36 - 20 - 24 + 12 \\ &= \mathbf{236}. \end{aligned}$$

Counting overview: Combine counts

First define one outcome. Then choose how to count it.

Principle	When it applies
Product rule	Build an outcome in stages; multiply when each stage has a fixed number of choices.
Sum rule	Split into disjoint cases; add their counts.
Complement	Count all candidates, then subtract the invalid ones.
Division rule	Each outcome has exactly d descriptions; divide by d .
Inclusion–exclusion	Overlapping cases: singles $-$ pairs $+$ triples $- \dots$.

Counting overview: Models and proof tools

Tool	Recognize the structure
Permutations	Order matters, no repetition: $P(n, r) = n!/(n - r)!$.
Combinations	Order does not matter, no repetition: $\binom{n}{r}$.
Stars and bars	r identical objects in n distinct boxes, allowing empty boxes: $\binom{r+n-1}{n-1}$.
Binomial theorem	Choosing k of n factors gives the coefficient $\binom{n}{k}$.
Combinatorial proof	Count the same outcomes in two ways, or give a bijection.
Pigeonhole principle	N objects in k boxes: some box has at least $\lceil N/k \rceil$ objects.

Check order, repetition, and overcounting before choosing a formula.